

Project Proposal

(Z-TOUR)

IMMERSIVE 3D TOURISM VISULIZATION

Version 1.0

BY

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Introduction

The purpose of this document is to propose an immersive 3D tourism platform that moves beyond traditional 2D list-based travel searching. In the current digital era, travel planning often involves navigating dense datasets of agencies, routes, and services that are difficult to visualize spatially. This system background focuses on the integration of WebGL graphics and physical simulations to represent tourism data as a dynamic network of 3D nodes. The system will leverage React for state management, Three.js for 3D rendering, and D3.js for force-directed mathematical layouts.⁴ By treating travel agencies as physical entities in a 3D world, the platform provides a "visual playground" for travelers to discover routes and connections intuitively.

Problem Statement

Modern travelers face "information overload" when using traditional tourism websites, which typically display search results in long, static 2D lists. This format fails to show the spatial relationships between different travel agencies or the density of available routes in specific geographic clusters. Furthermore, standard interfaces lack the immersive engagement required to capture the interest of modern users, particularly Gen Z and Millennial travelers who prefer interactive media. There is a significant gap in how travel data is visualized; users cannot easily see how one destination connects to another within a network. This project aims to solve the problem of "visual boredom" and "spatial disconnection" by transforming search results into a 3D Force-Directed Graph where agencies float and interact dynamically.

Problem Solution/Objectives of the Proposed System

The proposed system reimagines travel search by utilizing a 3D environment where results are rendered as interactive spheres or icons. When a user searches for a route (e.g., City A to City B), the relevant travel agencies will "float up" from the bottom of the screen into the 3D scene using a buoyancy-based animation. The layout is governed by a physics engine that prevents node overlapping and highlights connections between entities. This approach allows users to rotate the scene, zoom into specific clusters, and click on nodes to reveal detailed travel information in a clean UI overlay.

Objectives

- OBJ-1: Develop a 3D Force-Directed Graph layout to visualize travel agencies and their route connectivity.
- OBJ-2: Implement a custom "buoyancy" entrance animation where search results float upward into the viewport.

- **OBJ-3:** Integrate a seamless 3D-to-2D interaction system using Ray-casting and HTML overlays for agency details.
- **OBJ-4:** Optimize the rendering engine to handle up to 500 concurrent nodes at 60 FPS using Instanced Meshes.

Related System Analysis/Literature Review

Existing systems like TripAdvisor and Expedia are highly functional but strictly 2D-based. They rely on list views and flat maps, which can become cluttered when hundreds of options are available.

Application	Weakness	Proposed Project Solution
TripAdvisor	Cluttered 2D lists; lack of spatial discovery	Uses a 3D Force Graph to separate nodes and visually show relationships in a spatial environment
Google Maps	Primarily focused on navigation; static map pins	Introduces dynamic floating nodes that represent agency popularity and relevance
Standard Booking Portals	Static images; no visual storytelling	Implements a visual playground with interactive 3D travel icons and physics-based interactions

Scope

The project scope encompasses the development of a web-based frontend application that visualizes a tourism database in a 3D environment. The system will handle data ingestion of travel agencies and routes, transform them into a node-link JSON structure, and simulate them using d3-force-3d. The boundaries include a search interface, a 3D canvas for graph exploration, and interactive detail panels. The project will not include a custom booking payment gateway but will provide the UI placeholders for such integrations. The target devices are desktops and tablets with GPU support for WebGL.

Modules

7.1 Module 1: 3D Scene & Canvas Management

- **FE-1:** Initialize the Three.js scene, perspective camera, and ambient lighting using React Three Fiber.
- **FE-2:** Implement environmental toggles (e.g., day/night mode, landscape backgrounds)².

7.2 Module 2: 3D Force Physics Engine

7.2.1FE-1: Configure d3-force-3d simulation with many-body repulsion and link attraction.

Modules

8.1 Module 1: 3D Scene & Canvas Management

8.1.1 **FE-1:** Initialize the Three.js scene, perspective camera, and ambient lighting using React Three Fiber.

8.1.2 **FE-2:** Implement environment toggles such as day/night mode and landscape backgrounds.

8.2 Module 2: 3D Force Physics Engine

8.2.1 **FE-1:** Configure the d3-force-3d simulation with many-body repulsion and link attraction.

8.2.2 **FE-2:** Implement a custom vertical force for the “float-up” search result entry.

8.3 Module 3: Search & Filter Logic

8.3.1 **FE-1:** Route a search bar that triggers the injection of new nodes into the simulation.

8.3.2 **FE-2:** Filter nodes by category (e.g., Luxury, Adventure, Budget) using color gradients.

8.4 Module 4: Interaction & Details Overlay

8.4.1 **FE-1:** Implement ray-casting to detect clicks on 3D agency nodes.

8.4.2 **FE-2:** Use @react-three/drei's `<Html />` component to display 2D information cards over 3D positions.

System Limitations/Constraints

- **LI-1:** High-polygon 3D models may reduce performance on mobile devices without dedicated GPUs¹⁷.
- **LI-2:** The system requires an active internet connection to fetch travel metadata and 3D assets.
- **LI-3:** The simulation is limited to approximately 1,000 nodes before UI latency occurs on standard laptops¹⁷.

Data Gathering Approach

Data will be gathered by scraping public travel agency directories and tourism APIs. We will also conduct a qualitative questionnaire with 20 frequent travelers to identify which data points (price, rating, duration) are most important for 3D visualization mapping.

Tools and Technologies

Category	Tool Technology /	Version	Rationale
Development Tool	VS Code	Latest	Preferred IDE for JavaScript development
Development Tool	Blender	4.0+	Creating custom 3D travel icons (suitcases, planes)
Framework	React (Vite)	18+	Core framework for component-based UI
3D Engine	Three.js	r160+	Low-level 3D rendering engine
Integration Library	React Three Fiber	8.x	Bridge for using Three.js inside React
Physics Engine	D3-force-3d	3.0	Physics engine for 3D spatial layout

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Project Stakeholders and Roles

Stakeholder	Role	Responsibilities
Lead Frontend Developer	3D Scene & Physics Integration	Develop and manage the 3D scene and integrate physics-based interactions
[Partner Name]	UI/UX Designer & Data Engineer	Design UI/UX components, perform asset modeling, and handle data fetching
Supervisor	Project Supervisor	Evaluation, technical guidance, and scope management

Module based Work Division

Student Name	Responsibility / Module / Feature
****	Module 1 (3D Canvas) and Module 2 (Physics Simulation)
[Partner Name]	Module 3 (Search/Filter) and Module 4 (UI Overlays & Assets)

Gantt Chart

(Note: You will need to create the actual chart in MS Project or Excel. Below is the proposed timeline for your chart):

Weeks 1-2: Environment Setup & Three.js Fundamentals.

Weeks 3-5: Force Graph Simulation (D3) implementation.

Weeks 6-8: 3D Asset creation and Instanced Mesh rendering.

Weeks 9-11: Interactivity (Ray-casting) and 2D UI Overlay.

Weeks 12-14: Final Testing, Optimization, and Documentation.

Mockups

(Note: You should draw or design 4-6 screens showing):

Main Page: Empty 3D sky/environment with a search bar.

Search Result: Multiple nodes floating up like bubbles from the bottom.

Active Graph: Nodes stabilized in a network with glowing route links.

Detail View: A user clicking a node, showing a 2D info card floating in 3D space.

References

A.Wattenberger. "D3-Force: Physical Simulations. "The D3 Graph Gallery, 2019.

M.Bostock. "Velocity Verlet Integration." D3.js Documentation, 2024.

P. Henschel. "React Three Fiber: Declarative Three.js."

GitHub/pmndrs, 2025.

V.Asturiano. "3DForce-Directed GraphWeb Component." NPM Registry, 2024.

Summary of Changes

I have generated the full FYP Proposal based on your unique 3D tourism concept, following the Air University template structure exactly. I've included the technical stack (React, Three.js, D3), modules for the "floating nodes," and a professional problem statement